

Carolina Del Pilar Díaz

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Summary

AI/ML Engineer, Senior Data Scientist, and Applied Statistician (Economist) with 20+ years of experience advising global enterprises in Fintech, AgTech, and Healthcare. I specialize in architecting and deploying end-to-end AI products and solutions using Python, Machine Learning, Deep Learning, and Multi-Agent Systems (GenAI, RAG, MCP). My approaches are mathematically robust and hallucination-resistant, tailored for regulated environments. I have shipped autonomous AI applications built with custom-coded Python pipelines, combining Causal Inference, Econometrics, and MLOps on on-premises, hybrid, and cloud infrastructures (AWS, Azure) — always focused on data sovereignty and measurable ROI for clients such as ICBC, Aerolíneas Argentinas, and AB InBev. Core stack: Python, AI, Machine Learning, Deep Learning, Computer Vision, MLOps, Data Science.

Experience

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| <p>ITP Patagonia, Senior AI Engineer & Forward Deployed Engineer (FDE)</p> <ul style="list-style-type: none"> Multi-Agent Architecture & Cloud Orchestration: Architected a proprietary Multi-Agent Framework and shipped autonomous AI apps for top-tier financial institutions (Banco Galicia, ICBC). Designed model-agnostic architectures that dynamically switch between leading LLMs (OpenAI GPT, Google Gemini, Anthropic Claude, LLaMA, Mistral) via custom-coded LangGraph pipelines on on-premises and hybrid-cloud servers. Solved critical 'black box' failures by injecting rigorous statistical validation layers into the agent's reasoning loop, ensuring data sovereignty and output reliability. | <p>Buenos Aires, Argentina
June 2024 – present</p> |
| <p>ITP Patagonia - ICBC, Senior Data Scientist</p> <ul style="list-style-type: none"> Developed predictive models for payroll clients and mutual fund investment using AWS SageMaker, AutoML, Canvas, and Neural Network and XGBoost. Built an anomaly detection model using Neural Networks (AENN), Isolation Forest, and SHAP for interpretability using AZURE framework. Currently developing a fraud detection model with AWS Fraud Detector, leveraging Decision Trees, Neural Networks, Time Series Analysis, and Custom Fraud Prediction using AutoML. | <p>Buenos Aires, Argentina
June 2024 – July 2025</p> |
| <p>Nubiral S.A., Senior Data Scientist. COE, Commission of Expert. Generative Models.</p> <ul style="list-style-type: none"> Airline solution: Integrated AWS Rekognition, Lambda, S3, LangChain, and OpenAI GPT for Computer Vision and Generative AI, improving object detection accuracy at Aerolíneas Argentina. Developed a Post-Call Analytics solution using AWS services (Transcribe, Comprehend, Lambda, QuickSight) to analyze customer interactions. Implemented Azure Cognitive Search and Form Recognizer for efficient PDF document management at Banco Macro. | <p>Buenos Aires, Argentina
Oct 2023 – May 2024</p> |
| <p>Pi Data Strategy & Consulting, Semi-Senior Data Scientist, Statistician, Economist</p> <ul style="list-style-type: none"> Built end-to-end ML pipelines using Databricks and Apache Spark for large-scale data processing; developed regression, classification, and predictive models deployed on Azure Technology. Achieve EDA, Statistical analysis, regression models, classification models, neuronal network. Sales Forecast for georeferenced Dataset. Multinomial predictive models applied to the level of customer satisfaction. | <p>Córdoba, Argentina
Mar 2021 – Oct 2023</p> |

National Institute of Agriculture Technology (INTA), Lead Statistician / Head of Statistical Research Manfredi, Córdoba, Argentina
Jan 2012 – Dec 2021

- Designed and implemented experimental designs for all research projects, ensuring robust statistical methodology across multiple areas.
- Applied neural networks, deep learning, and computer vision for UAV and agricultural machinery, enhancing crop health monitoring.

Education

Postgraduate Master's Degree	National University of Córdoba , Applied Statistics	Córdoba, Argentina Jan 2006 – Dec 2008
Diploma	National University of Córdoba , Data Science Machine Learning and applications	Córdoba, Argentina Jan 2019 – Dec 2020
Bachelor's degree	National University of Córdoba , Economics	Córdoba, Argentina Jan 1993 – Dec 1998

Skills

Agentic & Generative AI: Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, AutoGen, Semantic Kernel, RAG, MCP, Prompt Engineering, Causal AI, Hallucination Mitigation

LLMs & Foundation Models: OpenAI GPT, Google Gemini, Anthropic Claude, LLaMA, Mistral, Mixtral, Cohere, Ollama, Hugging Face — model-agnostic multi-provider architectures

Machine Learning & Stats: Deep Learning, Computer Vision, Causal Inference, Bayesian Statistics, Econometrics, Anomaly Detection, Time Series, MLOps

Cloud & Big Data: AWS (SageMaker, Lambda, Rekognition, Fraud Detector), Azure AI Services, Snowflake, Databricks, Apache Spark, Python, R, SAS

Languages: Spanish (Native), English (Intermediate), Portuguese (Basic)